PREMIUM

Full-Length Mock Test

Mock 03

Ecology and Evolution

GATE EY

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Evolutionary Biology — Natural Selection]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.2 [1 Mark] [EASY] [Population Ecology — Metapopulations]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.3 [1 Mark] [EASY] [Community Ecology — Species Interactions]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.4 [1 Mark] [EASY] [Evolutionary Biology — Molecular Evolution]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.5 [1 Mark] [EASY] [Population Ecology — Metapopulations]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.6 [1 Mark] [EASY] [Community Ecology — Succession]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.7 [1 Mark] [EASY] [Evolutionary Biology — Speciation]

Natural selection is:

A) Random

B) Directional

C) Stochastic

D) Lamarckian

Q.8 [1 Mark] [EASY] [Population Ecology — Life Tables]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.9 [1 Mark] [EASY] [Community Ecology — Succession]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.10 [1 Mark] [EASY] [Evolutionary Biology — Molecular Evolution]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.11 [1 Mark] [EASY] [Population Ecology — r/K Selection]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.12 [1 Mark] [EASY] [Community Ecology — Succession]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.13 [2 Marks] [MODERATE] [Evolutionary Biology — Macroevolution]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.14 [2 Marks] [MODERATE] [Population Ecology — Metapopulations]

r-selected species have:

A) High K

B) High r

C) Low r

D) Stable populations

Q.15 [2 Marks] [MODERATE] [Community Ecology — Biodiversity]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.16 [2 Marks] [MODERATE] [Evolutionary Biology — Macroevolution]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.17 [2 Marks] [MODERATE] [Population Ecology — Metapopulations]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.18 [2 Marks] [MODERATE] [Community Ecology — Species Interactions]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.19 [2 Marks] [MODERATE] [Evolutionary Biology — Macroevolution]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.20 [2 Marks] [MODERATE] [Population Ecology — Metapopulations]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.21 [2 Marks] [MODERATE] [Community Ecology — Ecological Niches]

Natural selection is:

A) Random

B) Directional

C) Stochastic

D) Lamarckian

Q.22 [2 Marks] [MODERATE] [Evolutionary Biology — Natural Selection]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.23 [2 Marks] [MODERATE] [Population Ecology — Metapopulations]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.24 [2 Marks] [MODERATE] [Community Ecology — Succession]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.25 [2 Marks] [MODERATE] [Evolutionary Biology — Natural Selection]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.26 [2 Marks] [MODERATE] [Population Ecology — Population Growth]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.27 [2 Marks] [MODERATE] [Community Ecology — Biodiversity]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.28 [2 Marks] [MODERATE] [Evolutionary Biology — Macroevolution]

r-selected species have:

A) High K

B) High r

C) Low r

D) Stable populations

Q.29 [2 Marks] [MODERATE] [Population Ecology — Life Tables]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.30 [2 Marks] [MODERATE] [Community Ecology — Species Interactions]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.31 [2 Marks] [MODERATE] [Evolutionary Biology — Macroevolution]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.32 [2 Marks] [MODERATE] [Population Ecology — Population Growth]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.33 [2 Marks] [HARD] [Community Ecology — Biodiversity]

Natural selection is:

A) Random
B) Directional
C) Stochastic
D) Lamarckian

Q.34 [2 Marks] [HARD] [Evolutionary Biology — Molecular Evolution]

r-selected species have:

A) High K
B) High r
C) Low r
D) Stable populations

Q.35 [2 Marks] [HARD] [Population Ecology — Metapopulations]

Natural selection is:

A) Random

B) Directional

C) Stochastic

D) Lamarckian

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Community Ecology — Biodiversity]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Evolutionary Biology — Molecular Evolution]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Population Ecology — r/K Selection]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Community Ecology — Species Interactions]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Evolutionary Biology — Molecular Evolution]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Community Ecology — Species Interactions]

Standard calculation using basic formula:

Answer: ____________________

Q.52 [2 Marks] [HARD] [Evolutionary Biology — Macroevolution]

Standard calculation using basic formula:

Answer: ____________________

Q.53 [2 Marks] [HARD] [Population Ecology — Metapopulations]

Standard calculation using basic formula:

Answer: ____________________

Q.54 [2 Marks] [HARD] [Community Ecology — Ecological Niches]

Standard calculation using basic formula:

Answer: ____________________

Q.55 [2 Marks] [HARD] [Evolutionary Biology — Molecular Evolution]

Standard calculation using basic formula:

Answer: ____________________

Q.56 [2 Marks] [HARD] [Population Ecology — Life Tables]

Standard calculation using basic formula:

Answer: ____________________

Q.57 [2 Marks] [HARD] [Community Ecology — Biodiversity]

Standard calculation using basic formula:

Answer: ____________________

Q.58 [2 Marks] [HARD] [Evolutionary Biology — Natural Selection]

Standard calculation using basic formula:

Answer: ________________________

Q.59 [2 Marks] [HARD] [Population Ecology — Metapopulations]

Standard calculation using basic formula:

Answer: ____________________________

Q.60 [2 Marks] [HARD] [Community Ecology — Succession]

Standard calculation using basic formula:

Answer: ____________________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	**Q.2:**	B	**Q.3:**	B
Q.4:	B	**Q.5:**	B	**Q.6:**	B
Q.7:	B	**Q.8:**	B	**Q.9:**	B
Q.10:	B	**Q.11:**	B	**Q.12:**	B
Q.13:	B	**Q.14:**	B	**Q.15:**	B
Q.16:	B	**Q.17:**	B	**Q.18:**	B
Q.19:	B	**Q.20:**	B	**Q.21:**	B
Q.22:	B	**Q.23:**	B	**Q.24:**	B
Q.25:	B	**Q.26:**	B	**Q.27:**	B
Q.28:	B	**Q.29:**	B	**Q.30:**	B
Q.31:	B	**Q.32:**	B	**Q.33:**	B
Q.34:	B	**Q.35:**	B	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	10
Q.52:	10	**Q.53:**	10	**Q.54:**	10
Q.55:	10	**Q.56:**	10	**Q.57:**	10
Q.58:	10	**Q.59:**	10	**Q.60:**	10

Detailed Solutions

Question 1
[Evolutionary Biology — Natural Selection]

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 2
[Population Ecology — Metapopulations]

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 3
[Community Ecology — Species Interactions]

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 4
[Evolutionary Biology — Molecular Evolution]

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 5
[Population Ecology — Metapopulations]

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 6
[Community Ecology — Succession]

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 7
[Evolutionary Biology — Speciation]

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 8
[Population Ecology — Life Tables]

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 9
[Community Ecology — Succession]

Natural selection is:

Answer:

(B)

Natural selection is differential survival/reproduction — directional process.

Question 10 *[Evolutionary Biology — Molecular Evolution]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 11 *[Population Ecology — r/K Selection]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 12 *[Community Ecology — Succession]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 13 *[Evolutionary Biology — Macroevolution]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 14 *[Population Ecology — Metapopulations]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 15 *[Community Ecology — Biodiversity]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 16 *[Evolutionary Biology — Macroevolution]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 17 *[Population Ecology — Metapopulations]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 18 *[Community Ecology — Species Interactions]*

r-selected species have:

Answer:

(B)

r-selected: high reproductive rate, unstable environments.

Question 19 *[Evolutionary Biology — Macroevolution]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 20 *[Population Ecology — Metapopulations]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 21 *[Community Ecology — Ecological Niches]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 22 *[Evolutionary Biology — Natural Selection]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 23 *[Population Ecology — Metapopulations]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 24 *[Community Ecology — Succession]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 25 *[Evolutionary Biology — Natural Selection]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 26 *[Population Ecology — Population Growth]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 27 *[Community Ecology — Biodiversity]*

Natural selection is:

Answer:

(B)

Natural selection is differential survival/reproduction — directional process.

Question 28 *[Evolutionary Biology — Macroevolution]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 29 *[Population Ecology — Life Tables]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 30 *[Community Ecology — Species Interactions]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 31 *[Evolutionary Biology — Macroevolution]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 32 *[Population Ecology — Population Growth]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 33 *[Community Ecology — Biodiversity]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 34 *[Evolutionary Biology — Molecular Evolution]*

r-selected species have:

Answer: (B)

r-selected: high reproductive rate, unstable environments.

Question 35 *[Population Ecology — Metapopulations]*

Natural selection is:

Answer: (B)

Natural selection is differential survival/reproduction — directional process.

Question 36 *[Community Ecology — Biodiversity]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37 *[Evolutionary Biology — Molecular Evolution]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 *[Population Ecology — r/K Selection]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 *[Community Ecology — Species Interactions]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 *[Evolutionary Biology — Molecular Evolution]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 *[Population Ecology — Metapopulations]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 *[Community Ecology — Ecological Niches]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 *[Evolutionary Biology — Speciation]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 *[Population Ecology — r/K Selection]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 *[Community Ecology — Succession]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Evolutionary Biology — Macroevolution]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Population Ecology — Life Tables]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Community Ecology — Ecological Niches]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Evolutionary Biology — Molecular Evolution]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Population Ecology — Metapopulations]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Community Ecology — Species Interactions]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 *[Evolutionary Biology — Macroevolution]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 *[Population Ecology — Metapopulations]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 *[Community Ecology — Ecological Niches]*

Standard calculation using basic formula:

Answer:

(10)

Apply the appropriate formula for the given data.

Question 55 *[Evolutionary Biology — Molecular Evolution]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56 *[Population Ecology — Life Tables]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57 *[Community Ecology — Biodiversity]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58 *[Evolutionary Biology — Natural Selection]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59 *[Population Ecology — Metapopulations]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60 *[Community Ecology — Succession]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.